English

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025-2026)

Test Pattern

NEET (UG)

MAJOR

18-05-2026

PRE-MEDICAL : ENTHUSIAST, LEADER & ACHIEVER COURSE PHASE - ALL PHASE

Test Booklet Code

L1

This Booklet contains **22** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

Your Target is to secure Good Rank in Pre-Medical 2026

SUBJECT : PHYSICS

Topic : SYLLABUS 1 + 2

1. Potential energy of an object at any postion x is given by, $U = \frac{Bx^2}{A + \sqrt{x}}$, The dimensions of B are :

(1) $[M\,L^{-1/2}\,T^{-2}]$

(2) $[M\,L^{1/2}\,T^{-2}]$

(3) $[M\,L^{1/2}\,T^{-1}]$

(4) $[M\,L^{-1/2}\,T^{-1}]$

2. Percentage error in measuring the radius and mass of a solid sphere are 1% and 2% respectively. The error in measurement of moment of inertia (I) about its diameter is :- $\left(I = \frac{2MR^2}{5}\right)$

(1) 9% (2) 6%

(3) 5% (4) 4%

3. One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 20 divisions of vernier scale coincide with 18 divisions of main scale, the least count of the callipers is :-

(1) 0.02 cm

(2) 0.01 cm

(3) 0.001 cm

(4) 0.002 cm

4. The current flowing through a resistor 10.932 ohm is 4.25 amp. The potential difference is 46.461 volt. The potential in significant figures is :

(1) 46.461 V (2) 46.46 V

(3) 46.4 V (4) 46.5 V

5. If a car covers $2/5^{th}$ of the total distance with v_1 speed and $3/5^{th}$ distance with v_2 speed, then average speed is :-

(1) $\frac{1}{2}\sqrt{v_1 v_2}$ (2) $\frac{v_1 + v_2}{2}$

(3) $\frac{2v_1 v_2}{v_1 + v_2}$ (4) $\frac{5v_1 v_2}{3v_1 + 2v_2}$

6. Choose the wrong statement.

(1) Zero velocity of a particle does not necessarily mean that its acceleration is zero.

(2) Zero acceleration of a particle does not necessarily mean that its velocity is zero.

(3) If speed of a particle is constant, its acceleration must be zero.

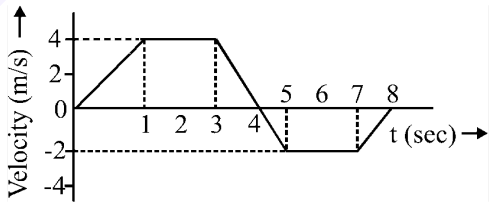
(4) None of the above.

7. A particle is moving in a straight line and passes through a point O with a velocity of $10\ ms^{-1}$. The particle moves with a constant retardation of $5\ ms^{-2}$ for 3 s and there after moves with constant velocity. How long after leaving O does the particle return to O ?

(1) 3 s (2) 8 s

(3) Never (4) 4.5 s

8. The velocity-time (v-t) graph of a linear motion of a body is shown in adjoining figure. The displacement of body from origin after 8 sec. is :-



(1) 6 meters

(2) 16 meters

(3) 8 meters

(4) 18 meters

9. A ball is thrown from the ground to clear a wall 3 m high at a distance of 6 m and falls 18 m away from the wall, the angle of projection of ball is

(1) $\tan^{-1}\left(\frac{3}{2}\right)$ (2) $\tan^{-1}\left(\frac{2}{3}\right)$

(3) $\tan^{-1}\left(\frac{1}{2}\right)$ (4) $\tan^{-1}\left(\frac{3}{4}\right)$

10. Rain drops are falling vertically downwards along $-y$ direction. A man is moving with a speed of 10 km/hr horizontally along $+x$ direction. He feels that rain drops are hitting him at an angle of 30° from vertical, then calculate velocity of rain drop with respect to man :-

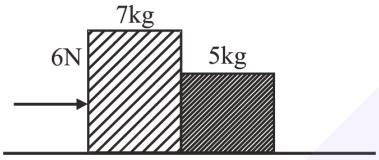
(1) $-10\hat{i} + 10\sqrt{3}\hat{j}$

(2) $10\hat{i} - 10\sqrt{3}\hat{j}$

(3) $10\hat{i} + 10\sqrt{3}\hat{j}$

(4) $-10\hat{i} - 10\sqrt{3}\hat{j}$

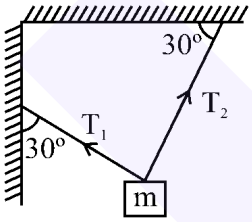
11. Two block of masses 7 kg and 5 kg are placed in contact with each other on a smooth surface. If a force of 6 N is applied on the heavier mass, the force on the lighter mass is :-



(1) 3.5 N (2) 2.5 N

(3) 7 N (4) 5 N

12. Calculate T_1 & T_2 :-



(1) $\frac{\sqrt{3}}{2}mg, \frac{mg}{2}$ (2) $mg, \frac{\sqrt{3}}{2}mg$

(3) $\frac{\sqrt{3}}{2}mg, \frac{\sqrt{3}}{2}mg$ (4) $\frac{mg}{2}, \frac{mg}{2}$

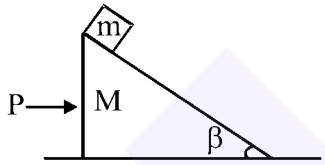
13. Two masses of 2 kg and 4 kg respectively are connected by a massless spring as shown in figure. A force of 60 N acts on the 4 kg mass at the instant when the 2 kg mass has an acceleration of 5 m/s^2 towards right, the acceleration of the 4 kg mass is :-



(1) 2.5 m/s^2 (2) 5.5 m/s^2

(3) 12.5 m/s^2 (4) 10.5 m/s^2

14. Two blocks are moving on a smooth horizontal surface such that the mass m remains stationary with respect to block of mass M as shown in the figure. The magnitude of force P is



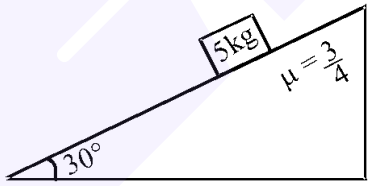
(1) $(M + m)g \tan \beta$

(2) $g \tan \beta$

(3) $mg \cos \beta$

(4) $(M + m)g \operatorname{cosec} \beta$

15. The force of friction acting on the block shown in figure is



(1) 30 N

(2) 25 N

(3) 20 N

(4) 15 N

16. A block of mass m is lying on an inclined plane. The coefficient of friction between the plane and the block is μ. The minimum force F required to move the block up the inclined plane will be–

(1) $mg \sin\theta + \mu mg \cos\theta$

(2) $mg \cos\theta - \mu mg \sin\theta$

(3) $mg \sin\theta - \mu mg \cos\theta$

(4) $mg \cos\theta + \mu mg \sin\theta$

17. A force $F = -5x$ Newton is acting on a particle and particle goes from $x = 5m$ to $x = 1m$, the work done by this force will be :

(1) +60 joule (2) –60 joule

(3) +75 joule (4) –75 joule

18. Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. Find the change in its kinetic energy between time t = 2sec and t = 4 sec :-

(1) 32 J (2) 46 J

(3) 61 J (4) 102 J

19. A body of mass 10 kg is released from a tower of height 20 m and body acquires a velocity of 10 ms^{-1} after falling through the distance 20 m. The work done by the push of the air on the body is :- (Take $g = 10\ m/s^2$)

(1) 1500 J (2) 1800 J

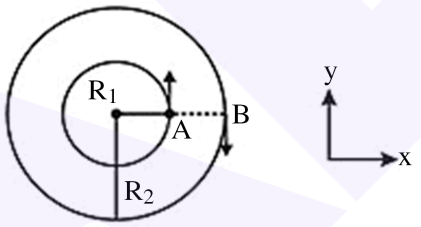
(3) −1500 J (4) −1800 J

20. A chain of mass m and length ℓ is held on a frictionless table in such a way that its $\frac{2\ell}{5}$ part is hanging below the edge of table. Find work done to pull the hanging part of chain.

(1) $\frac{2mg\ell}{25}$ (2) $\frac{mg\ell}{50}$

(3) $\frac{mg\ell}{25}$ (4) $\frac{4mg\ell}{25}$

21. In the figure given, A and B are two particles having same magnitude of angular velocity (ω) in opposite sense, Then $\overrightarrow{V_A} - \overrightarrow{V_B}$ at $t = \pi/2\omega$ sec is



(1) $-\omega R_1\hat{i} + \omega R_2\hat{i}$ (2) $\omega R_1\hat{i} - \omega R_2\hat{j}$

(3) $\omega R_1\hat{j} - \omega R_2\hat{i}$ (4) $\omega R_1\hat{j} + \omega R_2\hat{j}$

22. A stone of mass 1 kg is tied to the end of a string 1 m long. It is whirled in a vertical circle. If the velocity of the stone at the top be 4 ms^{-1}, what is the tension in the string at topmost point ? ($g = 10\ ms^{-2}$)

(1) 16 N (2) 10 N

(3) 6 N (4) 5 N

23. When a particle moves in a circle with uniform speed :-

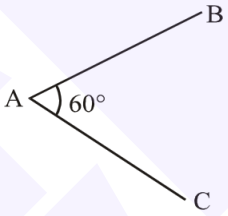
(1) its velocity and acceleration are both constant

(2) its velocity is constant but the acceleration changes

(3) its acceleration is constant but the velocity changes

(4) its velocity and acceleration both change

24. A uniform wire of length ℓ is bent into the shape of 'V' as shown. The distance of its centre of mass from the vertex A is



(1) $\ell/2$ (2) $\frac{\ell\sqrt{3}}{4}$

(3) $\frac{\ell\sqrt{3}}{8}$ (4) none of these

25. Two bodies of mass 10 kg and 2 kg are moving with velocity $(2\hat{i} - 7\hat{j} + 3\hat{k})$ m/s and $(-10\hat{i} + 35\hat{j} - 3\hat{k})$ m/s respectively. Find the velocity of their centre of mass.

(1) $\hat{k}$ m/s (2) $(2\hat{i} + 2\hat{k})$ m/s

(3) $\hat{j}$ m/s (4) $2\hat{k}$ m/s

26. A block of mass m moving at a velocity V collides with another block of mass 2m at rest. The block of mass m comes to rest after collision. Find the coefficient of restitution :-

(1) $\frac{1}{2}$ (2) 1 (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

27. A ball of mass m approaches a wall of mass M(>> m) with speed 4 m/s along the normal to the wall. The speed of wall is 1 m/s towards the ball. The speed of the ball after an elastic collision with the wall is :-

(1) 5 m/s away from the wall

(2) 9 m/s away from the wall

(3) 3 m/s away from the wall

(4) 6 m/s away from the wall

28. For a system to follow the law of conservation of linear momentum during a collision, the condition is :-
(i) total external force acting on the system is zero.
(ii) total external force acting on the system is non-zero.
(iii) total internal force acting on the system is zero.

(1) (i) only

(2) (ii) only

(3) (iii) only

(4) (i) or (ii)

29. A solid sphere and a hollow sphere of the same mass have the same moment of inertia about their respective diameters. The ratio of their radii will be :-

(1) 1 : 2 (2) $\sqrt{3} : \sqrt{5}$

(3) $\sqrt{5} : \sqrt{3}$ (4) 5 : 4

30. The torque of force $\vec{F} = -3\hat{i} + \hat{j} + 5\hat{k}$ acting on a point $\vec{r} = 7\hat{i} + 3\hat{j} + \hat{k}$ about origin will be :-

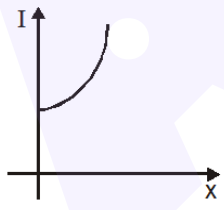
(1) $14\hat{i} - 38\hat{j} + 16\hat{k}$

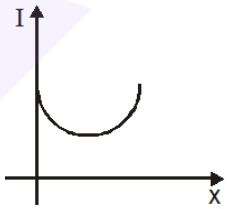
(2) $4\hat{i} + 4\hat{j} + 6\hat{k}$

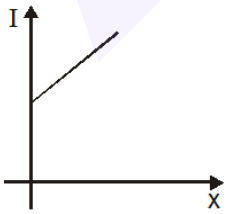
(3) $-14\hat{i} + 38\hat{j} - 16\hat{k}$

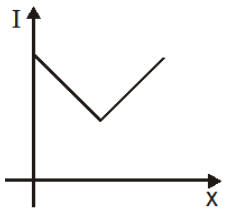
(4) $-21\hat{i} + 3\hat{j} + 5\hat{k}$

31. Moment of inertia I of a solid sphere about an axis parallel to a diameter and at a distance x from it varies as-

(1) 

(2) 

(3) 

(4) 

32. A homogeneous disc with a radius 0.2 m and mass 5 kg rotates about geometrical axis. The angular velocity of the rotation of the disc as a function of time is given by the formula $\omega = (2 + 6t)$ radian/sec. The tangential force applied to the rim of the disc is :-

(1) 1 N (2) 2 N (3) 3 N (4) 4 N

33. A rotating table completes one rotation in 10 sec. and its moment of inertia is 100 kg-m^2. A person of 50 kg mass stands at the centre of the rotating table. If the person moves 2m away from the centre, the angular velocity of the rotating table in rad/sec. will be:

(1) $\frac{2\pi}{30}$ (2) $\frac{20\pi}{30}$ (3) $\frac{2\pi}{5}$ (4) 2π

34. A solid sphere rolls down on an inclined plane of inclination θ. What will be the acceleration of the sphere ?

(1) $\frac{2}{7}g\sin\theta$ (2) $\frac{3}{5}g\sin\theta$

(3) $\frac{5}{7}g\sin\theta$ (4) $\frac{2}{5}g\sin\theta$

35. Which of the following is true for pure rolling on ground :-

(1) $V_{cm} = R\omega$ (2) $a_{cm} = R\alpha$

(3) Both (1) and (2) (4) None of these

36. What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of 2R ?

(1) $\frac{5GmM}{6R}$ (2) $\frac{2GmM}{3R}$

(3) $\frac{GmM}{2R}$ (4) $\frac{GmM}{3R}$

37. The escape velocity from a spherical planet surface is v. The escape velocity corresponding to another planet of twice the radius and half the mean density is :

(1) $\sqrt{2}v$ (2) $\frac{v}{2}$

(3) 2v (4) 4v

38. A particle of mass M is situated at the centre of a spherical shell of same mass and radius R. The gravitational potential at a point situated at $\frac{R}{2}$ distance from the centre will be -

(1) $-\frac{3GM}{R}$ (2) $-\frac{2GM}{R}$

(3) $-\frac{GM}{R}$ (4) $-\frac{4GM}{R}$

39. The radius of two planets are in the ratio 4 : 1 and their mean densities in the ratio 1 : 2. The acceleration due to gravity on the planets will be in ratio :-

(1) 1 : 2

(2) 2 : 3

(3) 4 : 1

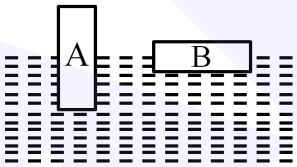
(4) 2 : 1

40. Pressure at the bottom of tank of water is 3P, where P is atmospheric pressure. If the water is drawn out it till the level of water is lowered by one fifth, then the pressure at the bottom of the tank is :-

(1) 2P (2) 13P/5

(3) 8P/5 (4) 4P/5

41. Two identical blocks of ice float in water as shown. Then :-



(1) Block A displaces a greater volume of water since the pressure acts on a smaller bottom area.

(2) Block B displaces a greater volume of water since the pressure is less on its bottom.

(3) The two blocks displace equal volumes of water since they have the same weight.

(4) Block A displaces a greater volume of water since its submerged end is lower in the water.

42. Two metal spheres are falling through a liquid of density 2×10^3 k/m^3 with the same uniform speed. The material density of sphere 1 and sphere 2 are 8×10^3 kg/m^3 and 11×10^3 kg/m^3 respectively. The ratio of their radii is :-

(1) $\frac{11}{8}$

(2) $\sqrt{\frac{11}{8}}$

(3) $\frac{3}{2}$

(4) $\sqrt{\frac{3}{2}}$

43. A water drop is divided into 8 equal droplets. The pressure difference between the inner and outer side of the big drop will be :

(1) same as for a smaller droplet

(2) 1/2 of that for a smaller droplet

(3) 1/4 of that for a smaller droplet

(4) twice of that for a smaller droplet

44. The total area of wings of an aeroplane is 15 m^2. The speed of air above and below the wings is 140 m/s and 110 m/s. Then the force on the aeroplane by air is
(Density of air = 1.28 kg/m^3)

(1) 48750 N

(2) 48000 N

(3) 95000 N

(4) 72000 N

45. If the work done in stretching a wire by 1 mm is 2 J, the work necessary for stretching another wire of same material but with double radius of cross-section and half the length by 1 mm is :-

(1) 16 J (2) 8 J

(3) 4 J (4) $\frac{1}{4}$ J

SUBJECT : CHEMISTRY

Topic : SYLLABUS 1 + 2

46. The angular momentum of electron in an excited H-atom is $\frac{2h}{\pi}$, the K.E. of electron will be:-

(1) + 13.6 eV (2) + 3.4 eV

(3) + 0.85 eV (4) + 0.54 eV

47. The uncertainties in the velocities of two particles, A and B are 0.05 and 0.02 ms^{-1}, respectively. The mass of B is five times to that of the mass of A. What is the ratio of uncertainties $\left(\frac{\Delta x_A}{\Delta x_B}\right)$ in their positions?

(1) 2 (2) 0.25 (3) 4 (4) 1

48. **Assertion :** The path of an electron in an atom can be determined accurately.

Reason : From the value of $|\psi|^2$ at different points within an atom, it is possible to predict the region around the nucleus where electron will be most probably found.

Choose the correct option :-

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect.

(4) Assertion is incorrect but reason is correct.

49. The orbital angular momentum of a 3p-electron is given as :-

(1) $\frac{3h}{2\pi}$ (2) $\sqrt{3}\frac{h}{2\pi}$ (3) $\frac{h}{\sqrt{2}\pi}$ (4) $\sqrt{\frac{3}{2}}\frac{h}{\pi}$

50. 8 g of A (mol. mass = 40) was placed in a 1L reaction vessel at 300 K and allowed to attain following equilibrium $A(g) \rightleftharpoons 2B(g)$. The total pressure at equilibrium was found to be 8 bar. K_p for the given reaction is approximately :

$\left(R = \frac{1}{12} L - atm\, mol^{-1} K^{-1}\right)$

(1) 18 bar (2) 3 bar

(3) 5 bar (4) 9.34 bar

51. For the manufacture of ammonia by the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 21.9$ k Cal, the favourable conditions are :-

(1) Low temperature, low pressure & catalyst

(2) Low temperature, high pressure & catalyst

(3) High temperature, low pressure & catalyst

(4) High temperature, high pressure & catalyst

52. Match the following columns

	Column-I [type of salt]		Column-II [relation btw K_{sp} and 's']
(i)	AgCl	(a)	$K_{sp} = 108\ s^5$
(ii)	$PbCl_2$	(b)	$K_{sp} = 27\ s^4$
(iii)	$AlCl_3$	(c)	$K_{sp} = s^2$
(iv)	$Ba_3(PO_4)_2$	(d)	$K_{sp} = 4\ s^3$

(1) (i)-(c), (ii)-(d), (iii)-(b), (iv)-(a)

(2) (i)-(a), (ii)-(d), (iii)-(b), (iv)-(c)

(3) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(a)

(4) (i)-(b), (ii)-(d), (iii)-(c), (iv)-(a)

53. The solubility of $BaSO_4$ in water is 10^{-5} M at 298 K. The solubility of $BaSO_4$ in 0.1 M K_2SO_4(aq.) solution is : (Molar mass of $BaSO_4$ = 233 g/mol)

(1) $10^{-9}\ gL^{-1}$

(2) $2.33 \times 10^{-7}\ gL^{-1}$

(3) $10^{-7}\ mol\ L^{-1}$

(4) $2.33 \times 10^{-7}\ mol\ L^{-1}$

54. Which of the following is a buffer solution?

(1) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.1 N NaOH

(2) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.1 N HCl

(3) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.2 N NaOH

(4) 500 mL of 0.2 N CH_3COOH + 500 mL of 0.1 N NaOH

55. A solution of 0.10 M NaZ has pH = 8.90. The K_a of HZ is :

(1) 1.6×10^{-4} (2) 1.6×10^{-5}

(3) 6.3×10^{-10} (4) 6.3×10^{-11}

56.

List-I (Amount of substance)		List-II (No. of moles of particular atoms in the given substance)	
(P)	6.022×10^{24} molecules of $Al_2(SO_4)_3.3H_2O$	(1)	15 mole O-atoms
(Q)	90 gm $C_6H_{12}O_6$	(2)	3 mole O-atoms
(R)	112 litre $SO_3(g)$ at 1 atm and 0°C	(3)	2.5 mole O-atoms
(S)	54 gram N_2O_5 (g)	(4)	150 mole O-atoms

Select the correct match

(1) P ⟶ 4;Q ⟶ 2;R ⟶ 1;S ⟶ 3

(2) P ⟶ 4;Q ⟶ 2;R ⟶ 3;S ⟶ 1

(3) P ⟶ 4;Q ⟶ 1;R ⟶ 2;S ⟶ 3

(4) P ⟶ 2;Q ⟶ 1;R ⟶ 4;S ⟶ 3

57. **Statement-I :** At STP 10 mL $CO_{2(g)}$ and 10 mL $CH_{4(g)}$ contain equal number of molecules.
Statement-II : Equal volumes of all gases at the same temperature and pressure contain equal number of molecules.

(1) Both statement-I and statement-II are true and statement-II is the correct explanation of statement-I.

(2) Both statement-I and statement-II are true but statement-II is not the correct explanation of statement-I.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true.

58. 50 g $CaCO_3$ will react with how many gram of 50% pure HCl sample according to the reaction $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

(1) 36.5 g (2) 73 g

(3) 18.75 g (4) 182.5 g

59. Which of the following statements are correct regarding a mixture containing 64.0 g H_2 and 64.0 g O_2 which is ignited so that water is formed as follows :-

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(\ell)$

I. H_2 is the limiting reagent

II. O_2 is the limiting reagent

III. After reaction, mixture contains 72.0 gm H_2O and 56.0 gm unreacted H_2 .

IV. After the reaction, mixture contains 56.0 gm H_2O and 72 gm unreacted H_2 .

(1) I and IV (2) II and IV

(3) II and III (4) I and III

60. Balance the following redox reaction and determine the value of x, y and z

$xCr^{+3} + yBrO^- + zOH^- \longrightarrow CrO_4^{-2} + Br^- + H_2O$

(1) 1, 3, 5 (2) 3, 1, 5

(3) 2, 3, 10 (4) 3, 2, 10

61. Consider the following compounds:
$\underline{K}O_2$, $H_2\underline{O}_2$ and $H_2\underline{S}O_4$.
The oxidation states of the underlined elements in them are, respectively,

(1) +1, –1, and +6 (2) +2, –2, and +6

(3) +1, –2, and +4 (4) +4, –4, and +6

62. 2 mole of an ideal gas expanded isothermally and reversibly from 1 litre to 10 litres at 300 K. Enthalpy change in the process would be :-

(1) 11.47 kJ (2) 4.98 kJ

(3) 0 kJ (4) – 11.40 kJ

63. Find out the value of equilibrium constant for the following reaction at 298 K.

$2NH_3(g) + CO_2(g) \rightleftharpoons NH_2CONH_2(aq) + H_2O(\ell)$

Standard Gibbs energy change $\Delta G°$ at given temperature is –13.6 kJ/mol (antilog of 0.38 = 2.4):-

(1) 2.4×10^3 (2) 4×10^2

(3) 2.4×10^2 (4) 2×10^3

64. What is the normal boiling point of mercury ?
Given :
ΔH_f° (Hg, ℓ) = 0; S°(Hg, ℓ) = 77.4 J/K-mol
ΔH_f° (Hg, g) = 60.8 kJ/mol; S°(Hg, g) = 174.4 J/K-mol

(1) 684.8 K (2) 626.8 K

(3) 536.8 K (4) None of these

65. $F_2C = CF - CF = CF_2 \longrightarrow \begin{matrix} F_2C - CF_2 \\ | \quad | \\ FC = CF \end{matrix}$

for this reaction (ring closure) $\Delta H = -49$ kJ mol^{-1}, $\Delta S = -40.2$ J K^{-1} mol^{-1}. Upto what temperature is the forward reaction spontaneous ?

(1) 1492° C (2) 1219° C

(3) 946° C (4) 1080° C

66. On the basis of the following thermochemical data :
$\left[\Delta H_f^0 \left(H^+_{(aq)} = 0\right)\right]$
$H_2O(\ell) \longrightarrow H^+(aq) + OH^-(aq)$; $\Delta H = 57.32$ kJ
$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$; $\Delta H = -286.20$ kJ
The value of enthalpy of formation of OH^- ion at 25°C is :-

(1) +228.88 kJ (2) −343.52 kJ

(3) −22.88 kJ (4) −228.88 kJ

67. For complete combustion of ethanol,
$C_2H_5OH(\ell) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$, the amount of heat produced as measured in bomb calorimeter, is 1364.47 kJ mol^{-1} at 25°C. Assuming ideality the Enthalpy of combustion, Δ_cH, for the reaction will be :-

(1) −1460.50 kJ mol^{-1}
(2) − 1350.50 kJ mol^{-1}
(3) − 1366.95 kJ mol^{-1}
(4) − 1361.95 kJ mol^{-1}

68. If the bond dissociation enthalpy of H_2, I_2 and HI are x, y and z kJ/mol respectively, then the enthalpy of formation of HI (in kJ) is-

(1) $x + 2y - 2z$ (2) $x + y - \frac{z}{2}$

(3) $\frac{x}{2} + \frac{z}{2} - y$ (4) $\frac{x}{2} + \frac{y}{2} - z$

69.

Atomic No. of Elements		Nature		Physical state	
(a)	55	(P)	Metal	(x)	Solid
(b)	31	(Q)	Non metal	(y)	Liquid
(c)	35	(R)	Metalloids	(z)	Gas
(d)	52	(S)	Transition Metal		

Correct match is :-

	a	b	c	d
1.	P, x	Q, y	Q, x	Q, x
2.	P, x	P, y	Q, y	Q, x
3.	P, y	P, y	Q, y	Q, x
4.	P, y	P, y	Q, y	R, x

70. In which of the following options order of arrangement agree with the variation of property indicated against it?
(A) $Al^{+3} < Mg^{+2} < Na^+ < F^-$ (Increasing ionic size)
(B) Be < C < N < F (Increasing non metallic character)
(C) Te < Se < O < S (Increasing negative electron gain enthalpy)
(D) Be < Mg < Ca < Sr (Increasing metallic radius)
Choose correct option.

(1) Only B and D (2) Only C

(3) Only B and C (4) Only A, B and D

71. Which one of the following statements is incorrect in relation to ionisation enthalpy?

(1) Ionisation enthalpy increases for each successive electron

(2) The greater increase in ionisation enthalpy is experienced on removal of electron from core noble gas configuration.

(3) Removal of valence electrons is marked by a big jump in ionisation enthalpy

(4) Removal of electron from orbitals bearing lower n value is easier than from an orbital having higher n value

72. Order of IInd I.P. of C, N, O, F is :

(1) C > N > O > F (2) O > N > F > C

(3) O > F > N > C (4) F > O > N > C

73. The correct order of electronegativities of following element will be-

(1) Li > Be > B > C > N

(2) C > S > I

(3) N > C > B > Be > Li

(4) C > Si > Pb

74. An Element having electronic configuration $1s^2$, $2s^2$, $2p^6$, $3s^2$,$3p^6$, $4s^1$ forms.

(1) Acidic oxide (2) Basic Oxide

(3) Amphoteric Oxide (4) Neutral Oxide

75. Wrong match is :-

(1) Z = 101 = Mendelivium

(2) Z = 106 = Seaborgium

(3) Z = 118 = Tennessine

(4) Z = 115 = Moscovium

76. Which of the following E.A. order is **not correct**?

(1) N < O < S (2) Cl > O > N > C

(3) O < S < F < Cl (4) I < Br < F < Cl

77. Choose incorrect option regarding Lewis dot structure ?

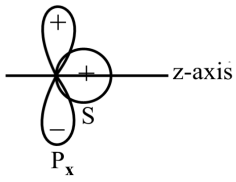
(1) The Lewis dot structure provide a picture of bonding in molecules and ions in terms of the shared pair electrons and the octet rule.

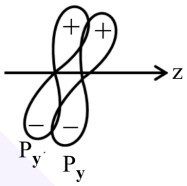
(2) The lowest energy structure is one with the smallest formal charge on the atoms.

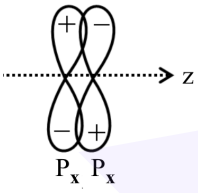
(3) The lowest energy structure is one with the highest formal charge on the atoms.

(4) The Lewis dot structure of CO_2 has equal no. of bond pair & lone pair.

78. Choose zero, +ve and –ve overlapping respectively from the following ?

(a) 

(b) 

(c) 

(1) a, b & c (2) b, c & a

(3) c, b & a (4) a, c & b

79. The hybridisation of central iodine atom in IF_5, I_3^- and I_3^+ are respectively :

(1) sp^3d^2, sp^3d, sp^3 (2) sp^3d, sp^3d, sp^3

(3) sp^3d^2, sp^3d^2, sp^3 (4) sp^3d, sp^3d^2, sp^3

80. Arrange the species according to increasing order of the no. of lone pair on their central atom from left to right :-

(1) XeF_2 , ICl_3 , SO_2 (2) ICl_3, SO_2 , XeF_2

(3) SO_2, ICl_3, XeF_2 (4) SO_2 , XeF_2 , ICl_3

81. Which of the following order is correct according to given property ?

(1) NH_2^- > NH_3 > NH_4^+ (Bond angle)

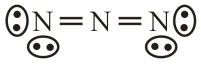
(2) O_3 > O_2 > H_2O_2 (Bond length)

(3) Cl_2 > Br_2 > F_2 > I_2 (Bond energy)

(4) C_2H_6 > C_2H_4 > C_2H_2 (Bond order)

82. What is the formal charge on 2, 1 & 3 atom respectively in given structure.

(1) (2) (3)



(1) – 1, +1, – 1 (2) + 1, –1, – 1

(3) – 1, –1, + 1 (4) + 1, +1, – 1

83. What is the correct order of S-O bond strength in following species SO_3, SO_4^{2-} & SO_3^{2-}

(1) $SO_4^{2-} > SO_3 > SO_3^{2-}$ (2) $SO_4^{2-} > SO_3^{2-} > SO_3$

(3) $SO_3 > SO_3^{2-} > SO_4^{2-}$ (4) $SO_3 > SO_4^{2-} > SO_3^{2-}$

84. Identify incorrect match

(1) PCl_3F_2 , nonplanar and nonpolar

(2) ClF_3 , planar and polar

(3) XeF_2 , non polar and planar

(4) XeF_6, nonpolar and planar

85. Correct dipole moment order is :-

(A) $CH_3Cl > CHCl_3$ (B) $CH_3Cl > CH_3F$

(C) $H_2O_2 > H_2O$ (D) $H_2O < NH_3 > NF_3$

(E) $SO_2 > SO_3$ (F) $HF > NF_3 > NH_3$

(1) A, B, C, E (2) A, B, C, F

(3) B,C,E,F (4) C, D, E, F

86. Which of the following is, correct order regarding the given property?

(1) AgF, AgCl, AgBr, AgI (decreasing order of covalent character)

(2) HF, HCl, HBr, HI (Decreasing order of pKa value)

(3) $AlCl_3$, NaCl, $BeCl_2$, LiCl (Increasing order of melting point)

(4) He, Ne, Ar, Kr, Xe (Decreasing order of Boiling point)

87. Given below are two statements :

Statement-I : The boiling point of hydrides of Group-16 elements follow the order $H_2O > H_2Te > H_2Se > H_2S$.

Statement-II : On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O, it has higher boiling point.

In the light of the above statements, choose the *correct* answer from the options given below :

(1) Both statement-I and Statement-II are true.

(2) Both statement-I and Statement-II are false.

(3) Statement-I is the true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

88. Match the column.

	Column-I (order)		Column-II (Properties)
(A)	$N_2 > N_2^+$	(P)	No. of σ bond
(B)	$B_2 > B_2^+$	(Q)	No. of unpaired e^-
(C)	$O_2^{2-} > O_2$	(R)	Bond order
(D)	$H_2 > H_2^+$	(S)	No. of A.B.M.O e^-s

(1) (A)-P,R (B)-Q,S (C)-S (D)-P,Q,R,S

(2) (A)-Q,R (B)-Q,R (C)-P,Q (D)-Q,R,S

(3) (A)-P,Q,R (B)-P,Q,R (C)-Q,R (D)-R,S

(4) (A)-P,R (B)-Q,R (C)-S (D)-P,R

89. Polarisation is the distortion of the shapes of an anion by an adjacently placed cation. Which of the following statements is correct ?

(1) Maximum polarisation is brought by a cation of high charge

(2) Minimum polarisation is brought by a cation of low radius

(3) A large cation is likely to bring about a large degree of polarisation

(4) Polarisation power of cation is less than that of anion

90. How many of the following species contain $p\pi-p\pi$ as well as $p\pi-d\pi$ bonds?

CO_2, XeO_4, $SOCl_2$, CO_2, SO_2, SO_2Cl_2, SO_3, $COCl_2$

(1) 6

(2) 4

(3) 2

(4) 5

SUBJECT : BIOLOGY

Topic : SYLLABUS 1 + 2

91. In a taxonomic hierarchy 'Genus' is placed between :

(1) Kingdom and Class (2) Class and Family

(3) Family and Order (4) Species and Family

92. There is a need to standarise the naming of living organisms such that a particular organism is known by the same name all over the world. This process is called _A_.

(1) A = Identification (2) A = Nomenclature

(3) A = Classification (4) A = Systematics

93. **Assertion (A) :** Methanogens, Halophiles & Thermoacidophiles live in rumen of cattle.

Reason (R) : Archaebacteria live in harsh habitats such as extremely salty areas, hot springs and marshy areas.

Choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

94. Which of the following statement is incorrect ?

(1) Viruses lack cellular structure.

(2) Viroids are composed of genetic material surrounded by a protein coat.

(3) Prions are infections agents consisting of abnormally folded proteins.

(4) Viroid's are smaller than viruses and lack a protein coat.

95. The zoospores of the class Phycomycetes of Fungi :

(1) Are endogenously produced inside sporangium.

(2) Are exogenously produced outside sporangium

(3) Lack flagella

(4) Are sexual spores

96. *Nostoc*, *Euglena* and *Puccinia* are belongs to, respectively :-

(1) Monera, Plantae and Fungi

(2) Protista, Protista and Fungi

(3) Monera, Protista and Fungi

(4) Fungi, Plantae and Fungi

97. Which group of organisms forms blooms in water bodies ?

(1) Archebacteria (2) Cyanobacteria

(3) Euglenoids (4) Slime moulds

98. Which of the following is an example of Deuteromycetes ?

(1) *Trichoderma* (2) *Puccinia*

(3) *Albugo* (4) *Neurospora*

99. Which protists lack a cell wall and show consumer nature ?

(1) *Euglena*, Protozoans

(2) Diatom, Dinoflagellates

(3) Protozoans, Diatom

(4) Slime mould, Dinoflagellate

100. Choose the correct match:

(1) *Mucor* - Found in aquatic habitat

(2) *Rhizopus* - Commonly called bread mould

(3) *Albugo* - Obligate parasite on wheat

(4) *Aspergillus* - Produce zygospore

101. Which of the following statement is incorrect ?

(1) Diatoms are the chief producers in the oceans.

(2) The cell wall of dinoflagellates has stiff cellulose plates on the outer surface.

(3) Majority of Euglenoids are fresh water organisms found in stagnant water.

(4) Slime moulds are autotrophic protists.

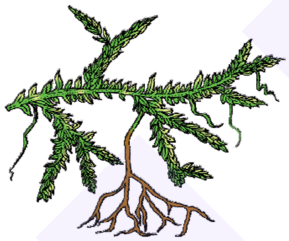
102.

	Column-I		Column-II
(A)	Slime mould	(i)	Forms massive plant body
(B)	Conidia	(ii)	The spore posses true walls
(C)	Virus	(iii)	Spirogyra
(D)	Oogamous	(iv)	Exogenously
		(v)	Infectious living fluid
		(vi)	Endogenously
		(vii)	*Volvox*, *Fucus*

(1) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)
(2) (A)-(ii), (B)-(iv), (C)-(v), (D)-(vii)
(3) (A)-(vi), (B)-(ii), (C)-(v), (D)-(vii)
(4) (A)-(vi), (B)-(ii), (C)-(v), (D)-(iii)

103. The sporophytic phase in *Funaria* is composed of :-
(1) Capsule only
(2) Seta and capsule only
(3) Foot and seta
(4) Foot, seta & capsule

104. Observe the following figure and select correct option.



(1) It represents *Sphagnum* belonging to mosses
(2) It represents *Selaginella* belonging to mosses
(3) It represents *Selaginella* belonging to lycopsida
(4) It represents *Sphagnum* belonging to mosses

105. Fusion between one large non-motile (static) female gamete and a smaller motile male gamete and gametes are formed in sex organ is termed as :-
(1) Anisogamy (2) Isogamy
(3) Oogamy (4) (1) and (2) both

106. Which of the following structure is not correctly matched with its ploidy level ?
(1) Spore - n (2) Zygote - 2n
(3) Sporophyte - n (4) Anthrozoids - n

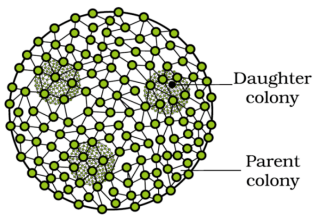
107. Match List-I with List-II and select the correct option :

	List-I		List-II
A	Agar-Agar	(i)	*Equisetum*
B	Gemma Cup	(ii)	*Gelidium* and *Gracilaria*
C	Strobili or Cones	(iii)	*Cycas*
D	Coralloid roots	(iv)	*Marchantia*

	A	B	C	D
1	ii	i	iv	iii
2	iii	ii	iv	i
3	ii	iv	i	iii
4	iv	ii	i	iii

108. Select the incorrect statements :
(a) In gymnosperms the male and female gametophytes remain within the sporangia retained on the sporophyte.
(b) The dicotyledon plants have trimerous flowers while monocotyledon plants have tetramerous or pentamerous flowers.
(c) The sex organs in bryophytes are unicellular.
(d) Being photosynthetic, the algae decrease the level of dissolved oxygen in their immediate environment.
(1) a and c (2) a, b and c
(3) b, c and d (4) a and d

109. Which of the following is true regarding the diagram given below :-



(1) Forms massive plant body
(2) Sexual reproduction is oogamous
(3) Used as food
(4) Produce large amount of hydrocolloids

110. In which stage of meiosis, reappearance of nucleolus occur ?

(1) Interkinesis (2) Telophase-1

(3) Anaphase-I (4) Metaphase-II

111. The diploid cell has 6 pg DNA in G_1 phase what would be the DNA content in this cell in G_2 - phase during cell cycle?

(1) 6 pg (2) 12 pg (3) 3 pg (4) 24 pg

112. Which of the following statements about metaphase is incorrect ?

(1) The complete disintegration of the nuclear envelope marks the start of metaphase.

(2) During metaphase, chromosomes are moved to the spindle equator and get aligned along the metaphase plate.

(3) Spindle fibres attach to kinetochores of chromosomes.

(4) During metaphase, chromosomes decondense and become invisible under the microscope.

113. Which one of the following stage of meiosis begins with the spliting of centromere?

(1) Anaphase (2) Metaphase II

(3) Anaphase II (4) Telophase II

114. Initiation of condensation of chromosomal material takes place in :-

(1) Interphase (2) Metaphase

(3) Prophase (4) Telophase

115. **Assertion :** In metaphase morphology of chromosome is most easily studied.
Reason : Condensation of chromosomes is completed by this stage.

(1) Both Assertion and Reason are True and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason is True but Reason is not the correct explanation of Assertion.

(3) Assertion is True but Reason is False.

(4) Both Assertion and Reason are False.

116. Match the Column-I and Column-II and select the correct option -

	Column-I		Column-II
(1)	Gap between meiosis-I and meiosis-II	(a)	Telophase-I
(2)	Dissolution of synaptonemal complex	(b)	Interkinesis
(3)	The nucleus of the daughter cell receives only one chromosome set from each homologous pair	(c)	Diplotene

(1) (1−b), (2−c), (3−a)

(2) (1−b), (2−a), (3−c)

(3) (1−a), (2−b), (3−c)

(4) (1−c), (2−b), (3−a)

117. What is the correct sequence of the steps given here ? Also work out the process depicted in the steps.
i. Homologous chromosomes move towards opposite poles of the cell
ii. Chromosomes gather at the two poles of the cell and the nuclear membranes reform
iii. Homologous chromosomes form pairs and exchange segments
iv. Homologous chromosomes align on equatorial plate

(1) The correct sequence is iii ⟶ iv ⟶ i ⟶ ii and the process is meiosis-I

(2) The correct sequence is ii ⟶ i ⟶ iv ⟶ iii and the process is mitosis

(3) The correct sequence is iv ⟶ i ⟶ iii ⟶ ii and the process is meiosis-I

(4) The correct sequence is iii ⟶ iv ⟶ i ⟶ ii and the process is meiosis-II

118. Considering the cell cycle which of the following statement is incorrect ?

(1) During the cell cycle DNA replication takes place

(2) Cell growth also takes place during cell cycle

(3) Cell cycle is a uncontrolled process

(4) Duplicated genome are distributed equally in daughter cells during mitosis

119. The starch-I_2 is _____ in colour. Cellulose does not contain complex_____ and hence cannot hold I_2 : Fill in the blanks, respectively :

(1) Red; Helices (2) Blue; Helices

(3) Red; Branches (4) Blue; Circle

120. What is odd one from the following ?

(1) Tyrosine (2) Palmitic acid

(3) Tryptophane (4) Phenyl alanine

121. Study the following statements carefully and select the correct statement.

(1) The enzymes, catalysing inter-conversion of optical, geometric or positional isomers are called isomerases

(2) Enzymes catalysing the linking together of two compounds are grouped in ligases.

(3) Enzymes catalysing hydrolysis of substrates included in hydrolases

(4) All of the above

122. Find out the incorrect statement.

(1) The protein portion of the enzymes is called, apoenzyme

(2) Catalytic activity is lost when Co-factor is removed from the enzyme.

(3) Co-factor do not play a crucial role in the catalytic activity of the enzyme.

(4) All of these

123. Match column-I with column-II and select the correct option given below :

Column-I		Column-II	
(A)	Endothermic	(i)	Difference between energy of substrate from that of transition state
(B)	Exothermic	(ii)	Transition state
(C)	ES complex	(iii)	Energy level of product is lower than substrate.
(D)	Activation energy	(iv)	Energy level of product is higher than substrate

(1) (A) iii (B) iv (C) ii (D) i (2) (A) iii (B) iv (C) i (D) ii

(3) (A) iv (B) iii (C) ii (D) i (4) (A) iv (B) iii (C) i (D) ii

124. Match the column-I with column-II and choose the correct option from the codes given below :-

	Column-I		Column-II
(A)	Collagen	(i)	Hormone
(B)	RuBisCO	(ii)	Enzyme
(C)	Trypsin	(iii)	Most abundant protein on the earth
(D)	Insulin	(iv)	Most abundant protein in animal body

(1) A-i, B-ii, C-iii, D-iv

(2) A-iv, B-iii, C-i, D-ii

(3) A-iv, B-iii, C-ii, D-i

(4) A-iii, B-iv, C-ii, D-i

125. Match the column I with column II

	Column-I (Secondary metabolites)		Column-II (Examples)
(A)	Drugs	(I)	Concanavalin A
(B)	Alkaloids	(II)	Abrin
(C)	Toxins	(III)	Codeine
(D)	Lectins	(IV)	Curcumin
		(V)	Vinblastin

Choose the correct answer from the options given below :

(1) A-IV, B-V, C-II,III, D-I

(2) A-IV,V, B-III, C-II, D-I

(3) A-IV,V, B-III, C-I, D-II

(4) A-III, B-IV,V, C-II, D-I

126. The main arena of different cellular activities in cell is :

(1) Microbodies (2) Vacuoles

(3) Cytoplasm (4) Rough ER

127. Plastids are involved in various metabolic processes, including :-

(1) Protein synthesis

(2) Lipid storage

(3) Carbohydrates synthesis

(4) All of the above

128. Fill in the blanks with correct option.

The interphase nucleus has highly extended and elaborate nucleoprotein fibres caused __a__, __b__ and one or more spherical bodies called __c__.

(1) (a) chromatin
(b) nuclear matrix
(c) nucleoli

(2) (a) chromatid
(b) nucleoli
(c) nuclear matrix

(3) (a) nucleoplasm
(b) nuclear matrix
(c) centromere

(4) (a) chromatid
(b) nucleoli
(c) kinetochore

129. Which pair of structures are usually found in both Plants and Fungi ?

(1) Cell wall and Chloroplast

(2) Cell membrane and Chloroplast

(3) Cell wall and Cell membrane

(4) Vacuoles and Chloroplast

130.

A.	Transport of hydrophobic molecules from higher concentration to lower concentration	(i)	Osmosis
B.	Transport of uncharged hydrophilic molecules from higher concentration to lower concentration	(ii)	simple diffusion
C.	Movement of water from higher to lower concentration of water across membrane.	(iii)	Active transport
D.	Transport of hydrophilic molecules against the concentration gradient.	(iv)	facilitated diffusion

Select the correct match for above table.

(1) A–iii, B–ii, C–iv, D–i

(2) A–iii, B–i, C–iv, D–ii

(3) A–i, B–ii, C–iv, D–iv

(4) A–ii, B–iv, C–i, D–iii

131. Which of the following statement is incorrect regarding mitochondria ?

(1) Mitochondria, unless specifically stained, are not easily visible under the microscope

(2) The number of mitochondria per cell is variable.

(3) The number of mitochondria depends on the physiological activity of cells.

(4) It is sausage-shaped or cylindrical having a diameter of 2.2-10.0 µm (average 5 µm) and length 1.0-4.1 µm.

132. Arrange the following cells in descending order of their size :

(1) Ostrich's egg ⟶ RBC of humans ⟶ Bacteria ⟶ *Mycoplasma*

(2) Ostrich's egg ⟶ *Mycoplasma* ⟶ RBC of humans ⟶ Bacteria

(3) *Mycoplasma* ⟶ Bacteria ⟶ RBC of humans ⟶ Ostrich's egg

(4) *Mycoplasma* ⟶ Bacteria ⟶ Ostrich's egg ⟶ RBC of humans

133. How many of the following statements are **incorrect**?

(A) Nucleus as a cell organelle was first described by Robert Brown as early as 1831.

(B) The interphase nucleus has highly condensed nucleoprotein fibres called chromatin.

(C) During different stages of cell division, cells show structured chromosomes in place of the nucleus.

(D) Every chromosome essentially has a primary constriction which is also known as centromere.

(E) Many membrane bound minute vesicles called microbodies that contain various enzymes are present only in plant cells.

(1) Two (2) One

(3) Three (4) Four

134. Match the columns :

	Column-I		Column-II		Column-III
(1)	*Mycoplasma*	(a)	Ostrich egg	(i)	Prokaryotic organism
(2)	Largest isolated single cell	(b)	52% Protein + 40% lipids	(ii)	Partially or totally buried
(3)	Membrane of human	(c)	Smallest cell	(iii)	For chemical studies on cell membrane
(4)	Membrane protein	(d)	Integral protein	(iv)	Eukaryotic cell

(1) (1)-(d)-(ii), (2)-(c)-(i), (3)-(a)-(iii), (4)-(b)-(iv)

(2) (1)-(c)-(i), (2)-(a)-(iv), (3)-(b)-(iii), (4)-(d)-(ii)

(3) (1)-(c)-(iv), (2)-(c)-(i), (3)-(b)-(ii), (4)-(b)-(iii)

(4) (1)-(c)-(iii), (2)-(a)-(iii), (3)-(b)-(i), (4)-(d)-(iv)

135. Arrange the following events of life cycle of bryophytes in correct sequence :-

I. Germination of spore

II. Gametes formation

III. Formation of Gametophyte

IV. Fertilization

V. Embryo formation

(1) I,III,IV,V,II

(2) II,IV,I,V,III

(3) III,II,IV,V,I

(4) V,I,II,III,IV

136. Which of the following pairs of animals belong same phylum?

(1) Pheretima, Cockroach

(2) Hydra, Sycon

(3) Fasciola, Ancylostoma

(4) Obelia, Aurelia

137. Vertebrata is a subphylum and Pisces is a

(1) Class (2) Super class

(3) Division (4) Sub class

138. Identify the Limbless amphibia.

(1) Hyla (2) Bufo

(3) Salamandra (4) Ichthyophis

139. Digestive system consists of both mouth and anus in which animals ?

(1) *Ancylostoma* (2) *Wuchereria*

(3) *Adamsia* (4) Both (1) and (2)

140. Name the second largest animal phylum.

(1) Porifera (2) Chordata

(3) Mollusca (4) Arthropoda

141. Given below are two statements one is labelled as assertion (A) and another is labelled as reason (R).

Assertion (A) : The members of phylum Aschelminthes are commonly known as round worms.

Reason (R) : The body of the members of this phylum appear circular in cross section.

In the light of the above statement choose the correct answer from the options given below.

(1) (A) is correct but (R) is not correct.

(2) Both (A) and (R) are correct but (R) is not correct explanation of (A)

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(4) Both (A) and (R) are incorrect.

142. Given below are two statements :

Statement-I : In open type circulatory system the blood is pumped out of the heart but the cells and tissues are not directly bathed in it.

Statement-II : In acoelomate animals the body cavity is lined by the mesoderm from all the sides.

In the light of the above statement choose the correct answer from the options given below.

(1) Statement-I and -II both are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Statement-I and II both are incorrect.

143. Identify the figure A, B, C and D and select the correct option :

(A)  (B) 

(C)  (D) 

	A	B	C	D
(1)	*Pleurobrachia*	*Cnidoblast*	*Aurelia*	*Adamsia*
(2)	*Aurelia*	*Adamsia*	*Cnidoblast*	*Pleurobrachia*
(3)	*Cnidoblast*	*Pleurobrachia*	*Adamsia*	*Aurelia*
(4)	*Adamsia*	*Aurelia*	*Pleurobrachia*	*Cnidoblast*

144. Choose the group of animals, which belong to different class.

(1) Chameleon, Hemidactylus, Cobra ⟶ Reptilia

(2) Icthyophis, Hyla, Rana, Bufo ⟶ Amphibia

(3) Struthio, Canis, Pteropus, Pterophyllum ⟶ Aves

(4) Equus, Macaca, Balaenoptera, Macropus ⟶ Mammalia

145. Match List-I with List-II.

	List-I		List-II
(a)	Ichthyophis	(i)	Cat
(b)	Cucumaria	(ii)	Rohu
(c)	Labeo	(iii)	Sea cucumber
(d)	Felis	(iv)	Limbless amphibia

(1) a-iv, b-iii, c-ii, d-i

(2) a-iv, b-iii, c-i, d-ii

(3) a-iv, b-i, c-iii, d-ii

(4) a-iv, b-ii, c-iii, d-i

146. Match the following column.

	I (Group)	II (Body feature)		III (Reproduction/ Character)		IV (Example)	
1	Pisees	A	Body covered with scales	W	Fertilization in water	i	Rohu
2	Amphibia	B	Moist glandular skin	X	Metamorphosis with tadpole stage	ii	Rana
3	Pisces	C	Gills present without opercalum	Y	Viviparous	iii	Cartilagineous fish
4	Amphibia	D	Three chambered heart	Z	Respiration by skin and lungs	iv	Toad

(1) 1-C-Y-iii, 2-D-W-iv, 3-A-X-i, 4-B-Z-ii

(2) 1-A-X-iv, 2-B-Z-ii, 3-C-W-iii, 4-D-Y-i

(3) 1-A-W-iii, 2-B-X-iv, 3-C-Z-i, 4-Y-D-ii

(4) 1-A-W-i, 2-B-X-ii, 3-C-Y-iii, 4-D-Z-iv

147. Reptiles are _____ animal. Fertilization is _____ and they are mostly _____.
Fill in the blanks with correct option :

(1) Diocious, internal, oviparous

(2) Bisexual, external, oviparous

(3) Unisexual, external, oviparous

(4) Monoecious, external, oviparous

148. All animals of which phylum are exclusively marine?

(1) Ctenophora

(2) Echinodermata

(3) Mollusca

(4) Both 1 and 2

149. Identify the non-chordate phylum of small marine worm-like organisms possessing an open circulatory system, gill for respiration, separate sexes with external fertilisation and indirect development.

(1) Neophron (2) Aplysia

(3) Calotes (4) Saccoglossus

150. Locomotion in Ctenophora is carried out by :-

(1) Jointed appendages
(2) Muscular foot
(3) Ciliated comb plate
(4) Parapodia

151. Which part of cockroach's alimentary canal is responsible for grinding of food ?

(1) Crop
(2) Ileum
(3) Proventriculus/Gizzard
(4) Hepatic caecae.

152. Which of the following statements is false about the circulatory system of cockroaches?

(1) Heart of cockroach is present along mid ventral line of thorax & abdomen.
(2) Blood vascular system of cockroach is open type.
(3) Blood from sinuses enter in heart through ostia and is pumped anterioly to next heart chamber.
(4) The haemolymph is composed of colourless plasma and haemocytes.

153. Choose correct sequence of food passing through the alimentary canal in frog.

(1) Mouth ⟶ Buccal cavity ⟶ Oesophagus ⟶ Pharynx ⟶ Stomach ⟶ Intestine
(2) Mouth ⟶ Buccal cavity ⟶ Pharynx ⟶ Stomach ⟶ Oesophagus ⟶ Intestine
(3) Mouth ⟶ Buccal cavity ⟶ Pharynx ⟶ Oesophagus ⟶ Stomach ⟶ First part of small intestine
(4) Mouth ⟶ Oesophagus ⟶ Buccal cavity ⟶ Stomach ⟶ Intestine

154. Find the incorrect statement regarding to from excretory system of frog.

(1) Kidney is dark red and bean shaped
(2) Ureter open into urinary bladder
(3) The thin walled urinary bladder is present at ventral to the rectum
(4) The frog excrete urea and thus is called ureotelic

155. In the brain of frog, optic lobes are present in the ___ ?

(1) Midbrain (2) Hindbrain
(3) Forbrain (4) Medulla oblongata

156. How many pairs of cranial nerves arise from the brain of frog ?

(1) 12 pairs (2) 10 pairs
(3) 8 pairs (4) 9 pairs

157.

Injury
↓
Platelets
↓
Thromboplastin
↓
(B)
↓
Prothrombin
↓
(C)
↓
Fibrinogen
↓
(D)

In reference to the above flow chart which statements are correctly given :
(i) Ca^{2+} plays a very important role in this process
(ii) 'D' is soluble in nature in which dead & damaged formed elements as trapped.
(iii) After completion of this process reddish brown scum formed at the site of a cut.
(iv) B-inactivates anticoagulant substance.

(1) (i), (ii), (iv) (2) (i), (ii), (iii)
(3) (ii), (iii), (iv) (4) (i), (iii), (iv)

158. **Statement-I :** A special neural centre in the pons can moderate the cardiac function through CNS.

Statement-II : Adrenal medullary hormones can increase the rate of heart beat and decrease the cardiac output.

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

159. **Assertion:** Platelets play a crucial role in the initiation of blood clotting.
Reason: Platelets release thromboplastin which helps in the conversion of fibrinogen into fibrin threads directly.
Choose the appropriate option:-

(1) Assertion and reason both are true and reason is correct explanation of Assertion.

(2) Assertion and reason both are true but reason is correct explanation of Assertion.

(3) Assertion is true but reason is false.

(4) Assertion and reason both are false.

160. Thrombocytes are cell fragments produced from ?

(1) Monocytes (2) Macrophages

(3) Megakaryocytes (4) Fibroblasts

161. Which one of the following mammalian cells is not capable of metabolism glucose to CO_2 by utilising O_2.

(1) RBC (2) WBC

(3) Unstraited muscle (4) Liver cells

162. The time duration from Lub to dub during a cardiac cycle is equal to time duration of :-

(1) Joint diastole

(2) Ventricular systole

(3) Ventricular diastole

(4) Atrial systole

163. A symptom of acute chest pain appears when no enough oxygen is reaching the heart muscle called as -

(1) Atherosclerosis (2) Angina

(3) Heart Failure (4) Hypertension

164. The cardiac output of an athlete will be much _____than that of an ordinary man.

(1) Lower (2) Higher

(3) Similar (4) Extremely lower

165. Match the following

Column 1	Column 2	Column 3
A. FRC	P. TV + IRV	I. 5800 ml
B. IC	Q. IC + ERV	II. 4600 ml
C. VC	R. TLC – IC	III. 3500 ml
D. TLC	S. VC + RV	IV. 2300 ml

(1) A-R-IV, B-P-III, C-Q-II, D-S-I

(2) A-R-IV, B-P-III, C-Q-I, D-S-II

(3) A-R-IV, B-Q-III, C-P-II, D-S-I

(4) A-R-IV, B-P-III, C-S-I, D-Q-II

166. Partial pressure of oxygen in alveoli, atmospheric air and tissue will be respectively :-

(1) 40, 159, 45 (2) 40, 0.3, 45

(3) 104, 159, 40 (4) 104, 0.3, 45

167. During -----A-----, ------B----- covered by a thin elastic cartilaginous flap called -----C---- to prevent the entry of food into the larynx. Here A, B and C are respectively.

(1) Gullet, Swallowing, Epiglottis

(2) Swallowing, Glutte, Epiglottis

(3) Swallowing, Glottis, Epiglottis

(4) Swallowing, Epiglottis, Glottis

168. What is the respiratory organ in adult frog ?

(1) Coxal gland (2) Lungs

(3) Gills (4) Tracheal Tube

169. How much amount of O_2 can deliver to tissue by 200 ml of oxygenated blood ?

(1) 5 ml (2) 1 ml (3) 2 lit (4) 10 ml.

170. Carotid body is sensitive for ?

(1) low CO_2 concentration

(2) low O_2 concentration

(3) High CO_2 concentration

(4) High O_2 concentration

171. What is Emphysema ?

(1) Shortness of breathing

(2) Inflammation of bronchi

(3) Alveolar walls damaged

(4) Hypoxea

172. In an average, a healthy human breathers _____ times/minute.

(1) 20-40 (2) 1-6 (3) 12-16 (4) 16-25

173. Which of the following statements is/are correct.
(a) nephrons are the functional units of the kidney
(b) Each kidney has 10 million nephrons
(c) Blood from the glomerulus is carried away by an afferent arteriole
(d) Glomerulus encloses the bowman's capsule.
(e) Glomerulus along with bowman's capsule, is called the malphighian body or renal carpuscle.

(1) four (2) three (3) two (4) one

174. Which of the following statements are not in correct ?
(a) Liver, the largest gland in our body secretes cholesterol, degrade steroid hormones and drugs.
(b) Sebaceous glands eliminate sebum, urea, NaCl.
(c) Sweat glands produce sweat which is oily fluid.
(d) In uremic patients urea can be removed by hemodialysis.

(1) a, b and c (2) a, b and d
(3) b and c (4) a and b

175. Match the column :

Column-I		Column-II	
A.	Glomerulonephritis	i.	Artificial kideny
B.	Haemodialysis	ii.	Kidney stones
C.	Renal calculi	iii.	Urea accmulation in blood
D.	Uremia	iv.	Inflammation glomerulus

(1) A-iii, B-i, C-ii, D-iv (2) A-iv, B-i, C-iii, D-ii
(3) A-iv, B-i, C-ii, D-iii (4) A-iv, B-ii, C-iii, D-i

176. Excretion in the form of uric acid and urates in birds is helpful in –

(1) Conservation of body heat
(2) Eliminating excess water
(3) Eliminating body water
(4) Conserving body water

177. A notch present on the medial side of kidney is known as –

(1) Ureter (2) Pelvis
(3) Hilus (4) Pyramid

178. What is the Excretory Structures present in Planaria ?

(1) Protonephridia
(2) Nephridia
(3) Flame cells
(4) Both (1) & (3)

179. In response to decrease in blood volume and blood pressure which of the following occur.

(1) Secretion of Renin
(2) Secretion of Aldosterone
(3) Secretion of ANF
(4) Secretion of Erythropoetin

180. What is Glomerular filtration Rate (GFR) ?

(1) 180 ml / minute
(2) 125 ml/minute
(3) 1250 ml/minute
(4) 300 ml/minute



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